

Macroeconomics II, Lecture VII:

Frictional labor markets: basics

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- Lectures I-VI: frameworks for business cycle analysis
- We learned a lot, but the models we considered had several problems and are obviously too stylized to speak to many macroeconomic phenomena
- In particular, two assumptions seem very unrealistic while also being important for understanding macro dynamics
 - ① Labor markets are frictionless
 - ② Households have access to complete asset markets (and can therefore be summarized by a representative agent)
- Remainder of the course introduces a set of models that relax these assumptions
- Besides developing a deeper theory of business cycle dynamics, these models will allow us to explore an additional set of questions in macro research:
 - ▶ How do macroeconomic conditions affect inequality? And how does inequality affect macroeconomic conditions?

- Lectures VII-X: Frictional labor markets
 - ▶ 4 lectures
 - ▶ Digging deeper into the determinants of household income
- Lectures XI-XIII: Incomplete asset markets
 - ▶ 3 lectures
 - ▶ Digging deeper into the determinants of consumption-savings dynamics, taking the income process as given
- To get us started with frictional labor markets, we'll first look at a few key facts

Today's agenda

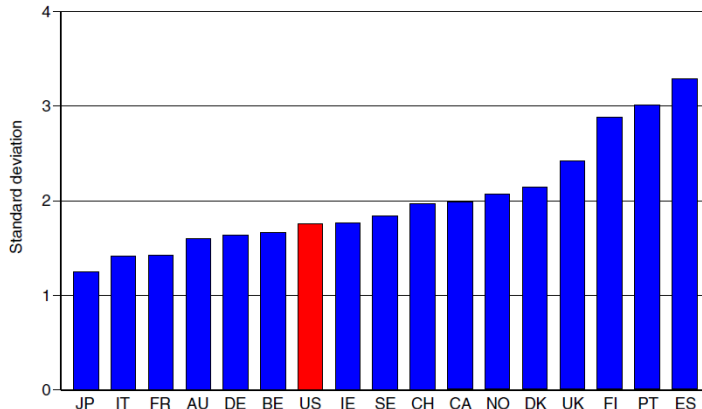
- ① Labor market facts
- ② Search models: overview
- ③ Mathematical Preliminaries
- ④ The McCall model

Labor market facts

- The labor market can be described in terms of
 - ▶ Stocks: hours worked, unemployment rate etc.
 - ▶ Flows: job-finding rate, separation rate etc.
 - ▶ Prices: wages
- We will look at some basic facts relating to each of these categories
- Main focus is the US, but we will also take a quick glance at other OECD countries
- Main data source: labor force surveys (US: CPS; Sweden: AKU)

- Main variables: Population N , Participation P , Hours worked H , Employed E , Unemployed U
- In some time interval Δt , a person is
 - ▶ **employed** if hours worked > 0
 - ▶ **unemployed** if not working, available for work and actively looking for work
 - ▶ **participating** if employed or unemployed
- Hours worked per capita $h^c = \frac{H}{N}$
- Question: how large are fluctuations in hours worked?

SD(hours per capita) cross countries



Standard deviation of detrended log points, using data 1965-2008. From Rogerson-Shimer (Handbook LE 2011).

- Fact 1: SD of cyclical fluctuations ~ 1.5 -2 percent for the US
- Fact 2: considerable heterogeneity in fluctuations across countries

Decomposition

- More definitions

- ▶ Hours worked per worker $h^w = \frac{H}{E}$

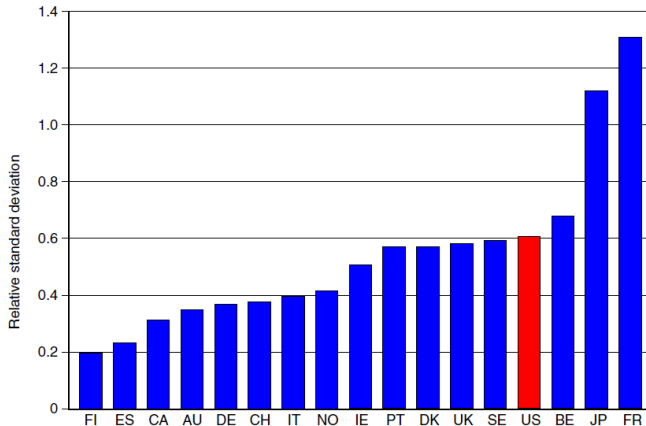
- ▶ Employment rate $e = \frac{E}{N}$

- Decomposing hours worked

$$\begin{aligned} h_t^c &= \frac{H_t}{N_t} \\ &= \frac{H_t}{E_t} \times \frac{E_t}{N_t} \\ &= \underbrace{h_t^w}_{\text{int. margin}} \times \underbrace{e_t}_{\text{ext. margin}} \end{aligned}$$

- Question: How much of fluctuations in hours worked per person is due to intensive vs extensive margin?

$SD(h_t^w)/SD(e_t)$ cross countries



Standard deviation of detrended log points, using data 1965-2008. From Rogerson-Shimer (Handbook LE 2011).

- Fact: for most countries, extensive margin is more important (but still considerable variation due to intensive margin)

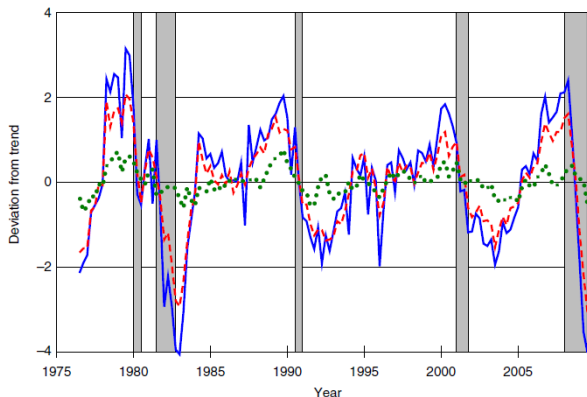
Further decomposition

- Even more definitions
 - ▶ Participation rate $p = \frac{P}{N}$
 - ▶ Unemployment rate $u = \frac{U}{P}$
- Decomposing hours worked

$$\begin{aligned}h_t^c &= \frac{H_t}{E_t} \times \frac{E_t}{N_t} \\&= h_t^w \times \frac{P_t - U_t}{N_t} \\&= h_t^w \times \frac{P_t}{N_t} \times \left(1 - \frac{U_t}{P_t}\right) \\&= h_t^w \times p_t \times (1 - u_t)\end{aligned}$$

- Question: How much of fluctuations in hours worked is due to participation vs. unemployment?

US time series

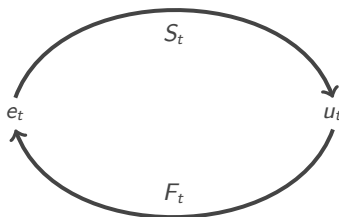


Detrended log points. Blue solid - hours worked per person; red dashed - employment rate; green dotted - participation rate. From Rogerson-Shimer (Handbook LE 2011).

- Fact: In the US, participation accounts only marginally for cyclical variation in hours worked
- $\Rightarrow$ most cyclical variation in hours worked is due to unemployment

Flows

- To gain further insight about the forces behind these fluctuations, we study the underlying flows
 - ▶ Participation fairly constant, so we restrict attention to flows in and out of unemployment
- Denote $F_t(\Delta t)$, $S_t(\Delta t)$ as the job-finding and job-separation probabilities in period $t \rightarrow t + \Delta t$
 - ▶ $F_t(\Delta t)$ = the fraction of unemployed flowing out of unemployment in period $t \rightarrow t + \Delta t$
 - ▶ $S_t(\Delta t)$ = the fraction of employed flowing into unemployment in period $t \rightarrow t + \Delta t$
- Graphical representation:



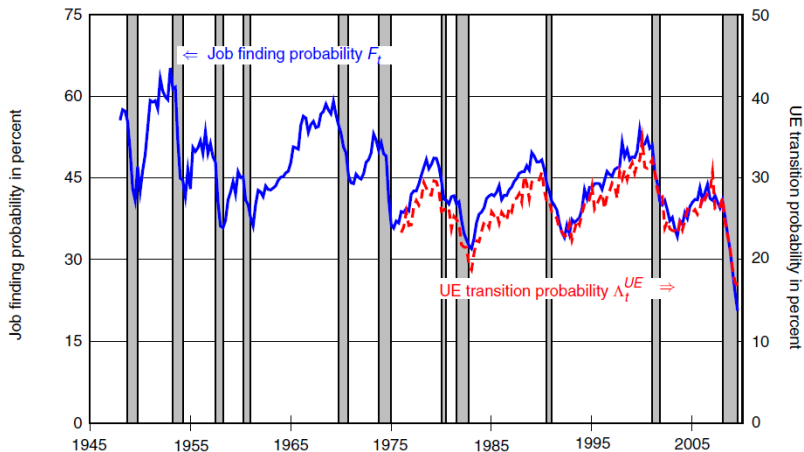
- Assume $\Delta t = 1$ month
- Using data on unemployment levels by duration, we estimate the probability F_t that an unemployed worker finds a job within a month:

$$1 - F_t = \frac{\# \text{Unemployed with duration} > 1 \text{ in month } t+1}{\# \text{Unemployed in month } t}$$

- Similarly, the monthly job-separation probability can be estimated from

$$S_t = \frac{\# \text{Unemployed with duration} < 1 \text{ in month } t+1}{\# \text{Employed in month } t}$$

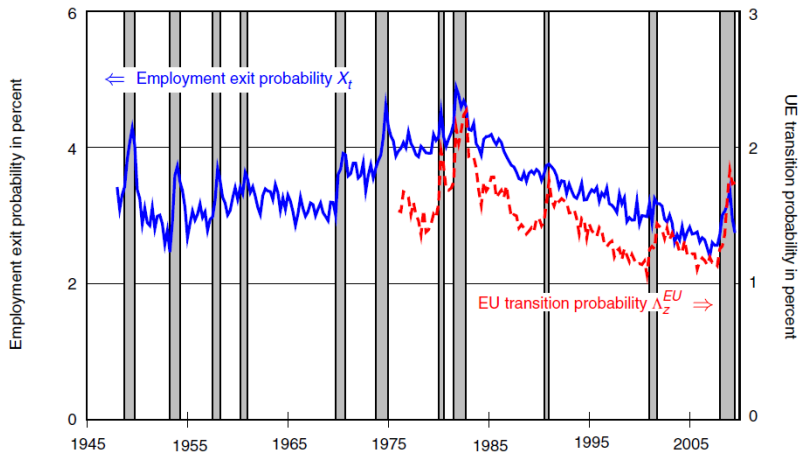
US job-finding probability



Blue solid line: inference from 2-state model; Red dashed line: inference from 3-state model. Rogerson-Shimer (Handbook LE 2011).

- Fact: Job-finding probability decreases (somewhat gradually) during recessions

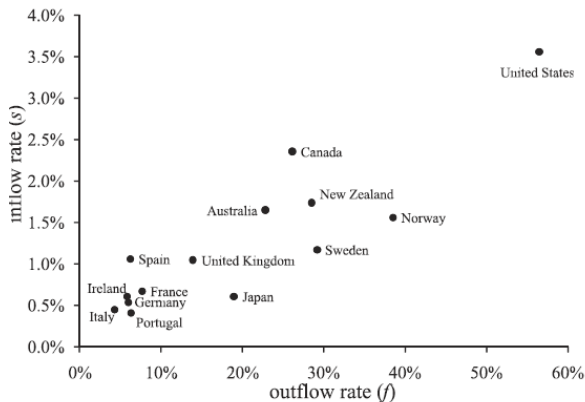
US job-separation probability



Blue solid line: inference from 2-state model; Red dashed line: inference from 3-state model. From Rogerson-Shimer (Handbook LE 2011).

- Fact: Job-separation probability spikes during recessions

Flows across countries

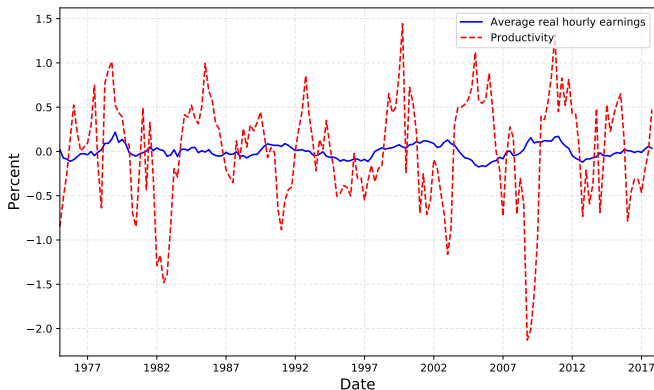


From Elsby-Hobijn-Sahin (ReStat 2013).

- Fact: US labor market is an outlier
- Fact: Strong positive association between job-separation and job-finding probability across countries

- The allocation of labor is coordinated via the labor price system, that is, the wage distribution
- How large are fluctuations in wages?
- Are different workers paid different wages?
- Are similar workers paid different wages?

US wage fluctuations



Detrended (HP-filter) quarterly data. OECD estimate of total labor productivity. Average hourly earnings for total private sector excluding supervisory employees, deflated with PCE. Source: FRED and own calculations.

- Fact: wage fluctuations $\ll$ productivity fluctuations
- Seemingly at odds with benchmark neoclassical model

US wage dispersion in CPS data

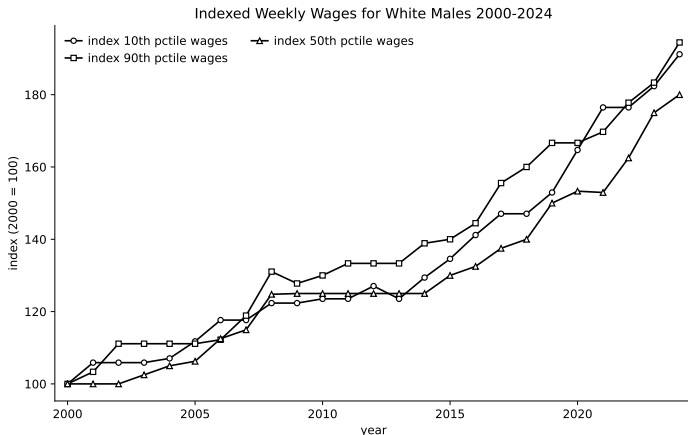


Indexed Wages for White Males 1963–97

From Acemoglu (JEL 2002).

- Fact: rising wage inequality 1975–2000

US wage dispersion: update through 2024 (using Cursor and GPT)



Own CPS ASEC calculations for white male full-time full-year workers aged 18-65, earnings years 2000-2024.

- (tentative) fact: Little action since 2000

Residual wage dispersion

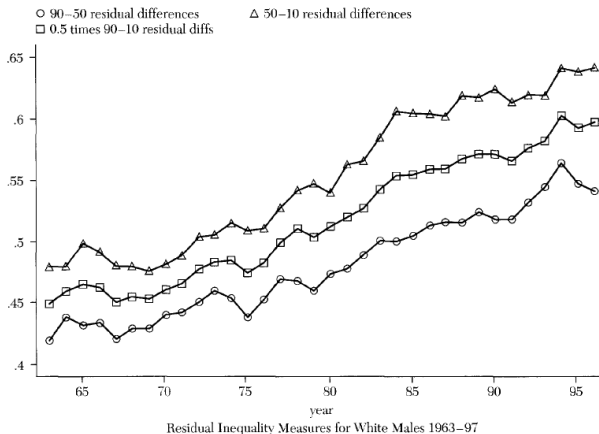
- How much of the observed wage dispersion is due to differences in observable characteristics?
- Cross-sectional household survey data (e.g. CPS, PSID)
- Mincer regression

$$\log y_{it} = \mu_y + \beta \mathbf{X}_{it} + \epsilon_{it}$$

where y_{it} is earnings and $\mathbf{X}_{it}$ includes age, occupation, industry, tenure, education, etc.

- Typically, $R^2 < 0.3$ (Mortensen, Book 2003)
- How to interpret the residual ϵ_{it} ?
 - ▶ Unobservable productivity differences? Luck?

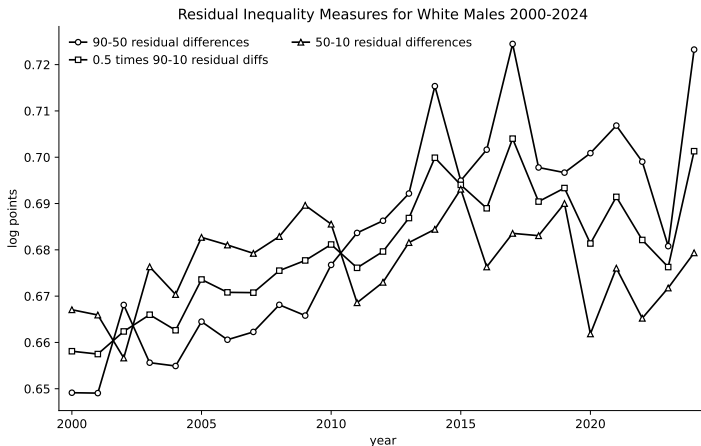
US residual wage dispersion in CPS data)



Y-axis: log points. From Acemoglu (JEL 2002).

- Fact: Rising residual wage inequality 1975–2000
- In 1995, median is $\sim e^{0.6} \approx 1.8$ times 10th percentile
- In 1995, 90th percentile is $\sim e^{2 \times 0.6} \approx 3.3$ times 10th percentile

US residual wage dispersion: update through 2024 (using Cursor and GPT)



Y-axis: log points. Own CPS ASEC calculations for white male full-time full-year workers aged 18-65, earnings years 2000-2024.

- (tentative) fact: Little action since 2000

Search models: overview

Search models

- Search models provide a theory of the labor market that links stock, flows and the distribution of wages together
- Search markets: markets in which buyers (employers) and sellers (workers) spend time/resources searching for each other, and where there is some element of randomness in when and/or with whom a match happens
- Because searching takes time, some workers will experience periods without work: **frictional unemployment**
- Because matching has an element of randomness, some workers will be luckier than others in finding good jobs: **frictional wage dispersion**
- Contrast to Walrasian markets (e.g. standard RBC and NK models)
 - ▶ Markets clear instantaneously
 - ▶ Unemployment can only arise if wages are not clearing the market
 - ▶ Law of one price holds: no wage dispersion among identical workers

- The search approach to labor markets was developed in the 60's, first largely focused on partial-equilibrium search behavior and later built into general-equilibrium frameworks
- We will cover
 - ▶ Lecture VII: Partial-equilibrium search (McCall)
 - ▶ Lecture VIII: Job-ladders and wage dispersion (Burdett-Mortensen)
 - ▶ Lecture IX-X: Unemployment (Diamond-Mortensen-Pissarides)

Mathematical Preliminaries

Math preliminaries I: Expected value

- For a non-negative random variable x with cdf $F(x)$ on support $[x_1, x_2]$, we can define the expected value as

$$Ex = \int_{x_1}^{x_2} x dF(x)$$

- Contrast with the “standard” definition

$$Ex = \int_{x_1}^{x_2} x f(x) dx$$

- For distributions where F is differentiable, these definitions coincide.
 - ▶ $\frac{dF(x)}{dx} = f(x) \Leftrightarrow dF(x) = f(x)dx$
 - ▶ This includes the usual suspects: Uniform, Normal, Frechet, Pareto, etc...
- However, when F has mass points, the expected value is not well defined under the latter definition (Draw example on whiteboard)
- Hence, we will employ the former definition

Math preliminaries II: Riemann-Stieltjes integrals

- Consider our definition of expected value

$$Ex = \int_{x_1}^{x_2} x dF(x)$$

- When the integrator is a function rather than a variable, we call it a **Riemann-Stieltjes integral**
- Properties of RS integrals:
 - 1 If F is differentiable: $\int G(x) dF(x) = \int G(x) f(x) dx$
 - 2 Integration by parts: $\int_a^b G(x) dF(x) = G(x)F(x)|_a^b - \int_a^b F(x) dG(x)$
 - 3 In particular: $\int_a^b dF(x) = F(x)|_a^b$
- (Compute Ex of example distribution on whiteboard)

Math preliminaries III: Two useful formulas for real analysis

- **l'Hôpital's rule:** if $f(x)$ and $g(x)$ are two differentiable functions (in an open interval around c) and

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0 \text{ or } \pm \infty$$

then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

- **Leibniz' rule:** If the functions $f(x, t)$, $\alpha(t)$, $\beta(t)$ are differentiable in t , the function

$$\phi(t) = \int_{\alpha(t)}^{\beta(t)} f(x, t) dg(x)$$

is differentiable and

$$\phi'(t) = f(\beta(t), t) \frac{dg(\beta(t))}{dt} - f(\alpha(t), t) \frac{dg(\alpha(t))}{dt} + \int_{\alpha(t)}^{\beta(t)} f_t(x, t) dg(x)$$

The McCall model

The McCall (QJE 1970) model

- Framework for thinking about optimal search behavior
- Identical households who search for employment opportunities
- Partial equilibrium: we take prices (and firm behavior) as given

- ① Model setup
- ② Reformulation of model in continuous time
- ③ Solution: reservation wage strategy
- ④ Implications:
 - ▶ Match quality
 - ▶ Unemployment duration
 - ▶ Unemployment dynamics
 - ▶ Wage dispersion
- ⑤ The Rothschild critique and the Diamond paradox

McCall assumptions

- An economy populated by a measure P of ex ante identical, infinitely lived agents with a linear utility function and discount factor β
- Households can either be **employed** or **unemployed**
- An **unemployed** household receives benefits b , while an **employed** household receives wage income w
- No savings technology, agents consume hand-to-mouth
- Unemployed households receive a job offer with probability Λ_u in each period, with wages w drawn from i.i.d. $F(w)$ with finite support $\mathbb{W} = [w_{\min}, w_{\max}]$
- Employed households separate from jobs at exogenous probability S
- Household problem: if unemployed and receiving a job offer, accept it or wait

- Search models are generally easier to work with in [recursive formulation](#)
- Value (i.e. Present-Discounted Value) of employed agent:

$$W(w) = w + \beta [(1 - S)W(w) + SU]$$

- Value of unemployed agent:

$$U = b + \beta \left[(1 - \Lambda_u)U + \Lambda_u \int_{\mathbb{W}} \max\{W(w), U\} dF(w) \right]$$

Value functions in continuous time

- Given these Bellman equations, we could proceed to solve for the worker decision and derive all results
- However, many macro-search models are formulated in continuous time
- We will therefore restate the previous problems in continuous time and work with this formulation from now on
- Same setup, but we denote the time period by Δt (and do not normalize it to 1)
- For given Δt , we distinguish between **rates** and **levels**
 - ▶ Job-finding rate λ_u ; Job-finding probability $\Lambda_u(\Delta t) = \lambda_u \Delta t$
 - ▶ Job-separation rate σ ; Job-separation probability $S(\Delta t) = \sigma \Delta t$
 - ▶ Discount rate r ; Discount factor $\beta(\Delta t) = e^{-r\Delta t} \approx \frac{1}{1+r\Delta t}$

Employed value function in continuous time

- PDV of employed agent

$$W(w) = w\Delta t + e^{-r\Delta t} [(1 - \sigma\Delta t)W(w) + \sigma\Delta tU]$$

- Rearrange

$$\frac{1 - e^{-r\Delta t}}{\Delta t} W(w) = w + \sigma e^{-r\Delta t} (U - W(w))$$

- l'Hôpital's rule says

$$\lim_{\Delta t \rightarrow 0} \frac{1 - e^{-r\Delta t}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{re^{-r\Delta t}}{1} = r$$

- so in the limit $\Delta t \rightarrow 0$:

$$rW(w) = w + \sigma(U - W(w))$$

- flow value rW = flow benefit + flow probability of a shock $\times$ change in value

Unemployed value function in continuous time

- PDV of unemployed agent:

$$\begin{aligned}U &= b\Delta t + e^{-r\Delta t} \left[(1 - \lambda_u\Delta t)U + \lambda_u\Delta t \int_{\mathbb{W}} \max\{W(w), U\}dF(w) \right] \\&= b\Delta t + e^{-r\Delta t} \left[U + \lambda_u\Delta t \int_{\mathbb{W}} \max\{W(w) - U, 0\}dF(w) \right]\end{aligned}$$

- Rearrange

$$\frac{1 - e^{-r\Delta t}}{\Delta t} U = b + e^{-r\Delta t} \lambda_u \int_{\mathbb{W}} \max\{W(w) - U, 0\}dF(w)$$

- Take the limit and use l'Hôpital's rule to find

$$rU = b + \lambda_u \int_{\mathbb{W}} \max\{W(w) - U, 0\}dF(w)$$

- Our problem is to find U , W and a wage acceptance rule $a : \mathbb{W} \rightarrow \{0, 1\}$ that solves our system of two equations

$$\begin{aligned}rW(w) &= w + \sigma(U - W(w)) \\ rU &= b + \lambda_u \int_{\mathbb{W}} \max\{W(w) - U, 0\} dF(w)\end{aligned}$$

- That's it
- Now we 1) solve this problem and 2) study the implications of the solution

Solution I: the reservation wage strategy

- We solve for W in terms of U :

$$rW(w) = w + \sigma(U - W(w)) \Rightarrow W(w) = \frac{w + \sigma U}{r + \sigma}$$

- Interpretation?
- Using this, we rewrite unemployed value

$$\begin{aligned} rU &= b + \lambda_u \int_{\mathbb{W}} \max\left\{\frac{w + \sigma U}{r + \sigma} - U, 0\right\} dF(w) \\ &= b + \lambda_u \int_{\mathbb{W}} \max\left\{\frac{w - rU}{r + \sigma}, 0\right\} dF(w) \end{aligned}$$

- $\frac{w - rU}{r + \sigma}$ increasing, 0 is a constant (Draw graph on whiteboard)
- Ergo, the wage acceptance rule is a reservation wage strategy: accept all wages $w \geq w_R$
- First testable hypothesis: households use reservation wage strategies

Solution II: characterizing the reservation wage strategy

- Because of the reservation strategy, the unemployed value can be rewritten

$$\begin{aligned} rU &= b + \lambda_u \int_{w < w_R} 0 dF(w) + \lambda_u \int_{w \geq w_R} \left(\frac{w - rU}{r + \sigma} \right) dF(w) \\ &= b + \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - rU) dF(w) \end{aligned}$$

- The reservation wage satisfies $w_R = rU$:

$$w_R - b = \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - w_R) dF(w)$$

- We call this the **reservation wage equation**
 - ▶ LHS = marginal cost of waiting
 - ▶ RHS = marginal gain of waiting
 - ▶ The equation solves for w_R
- Why is $w_R > b$?

Solution retrieved, what now?

- Reminder: the problem was to find the acceptance rule, which we have shown to be a cutoff strategy with reservation wage w_R
- This is all there is to it
- Solving explicitly for w_R requires knowledge about F
- We will remain agnostic about F here, but we can still derive a lot of properties and implications of the solution

Comparative statics I

- How does w_R vary with b ? Our solution satisfies

$$w_R - b = T(w_R) \equiv \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - w_R) dF(w)$$

- Differentiate w.r.t. b :

$$\frac{dw_R}{db} - 1 = T'(w_R) \frac{dw_R}{db}$$

- Rearrange:

$$\frac{dw_R}{db} = \frac{1}{1 - T'(w_R)}$$

- Apply Leibniz's rule:

$$\begin{aligned} T'(w_R) &= \frac{\lambda_u}{r + \sigma} \left[(w_{\max} - w_R) \frac{\partial F(w_{\max})}{\partial w_R} \right. \\ &\quad \left. - (w_R - w_R) \frac{\partial F(w_R)}{\partial w_R} + \int_{w \geq w_R} -1 dF(w) \right] \\ &= -\frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} dF(w) \\ &< 0 \end{aligned}$$

- Ergo, the reservation wage increases with unemployment benefits

- Similarly, one can show

$$\frac{dw_R}{d\lambda_u} > 0, \quad \frac{dw_R}{dr} < 0, \quad \frac{dw_R}{d\sigma} < 0$$

- Intuition?

- What about changes in F ?
- A uniform increase in $F(w)$ increases the reservation wage
- How does w_R respond to an increase in dispersion?
 - ▶ see your problem set

Match quality

- The observed wage distribution is $G(w; w_R) = F(w|w > w_R)$
- An increase in w_R implies that observed wages are higher
- More precisely: if $w_R^1 > w_R^2$, then $G_1(w; w_R^1)$ first-order stochastically dominates $G_2(w; w_R^2)$
 - ▶ The only difference between $G_1(w; w_R^1)$ and $G_1(w; w_R^2)$ is that lower wage offers are discarded in $G_1(w; w_R^1)$
- Hypothesis: Everything that increases w_R (e.g. benefits b) will also increase observed wages
- Reformulated: higher unemployment benefits should increase observed match quality

Unemployment dynamics

- The economy is populated by P workers
- In a time period Δt , a fraction $\sigma\Delta t$ of employed workers $E_t = P - U_t$ separate and a fraction $\lambda\Delta t$ of unemployed workers U_t find a job
 - ▶ $\lambda = \lambda_u(1 - F(w_R))$

- Law of motion for unemployment

$$U_{t+\Delta t} = U_t + \sigma\Delta t(P - U_t) - \lambda\Delta tU_t$$

- Divide by P :

$$u_{t+\Delta t} = u_t + \sigma\Delta t(1 - u_t) - \lambda\Delta tu_t$$

- Rearrange:

$$\frac{u_{t+\Delta t} - u_t}{\Delta t} = \sigma(1 - u_t) - \lambda u_t$$

- Take the limit $\Delta t \rightarrow 0$:

$$\dot{u}_t = \sigma(1 - u_t) - \lambda u_t$$

- Law of motion for unemployment:

$$\dot{u}_t = \sigma(1 - u_t) - \lambda u_t$$

- Given σ, λ , this ordinary differential equation has a unique solution converging to the steady-state value u .
- Steady state u defined by $\dot{u}_t = 0$:

$$u = \frac{\sigma}{\sigma + \lambda}$$

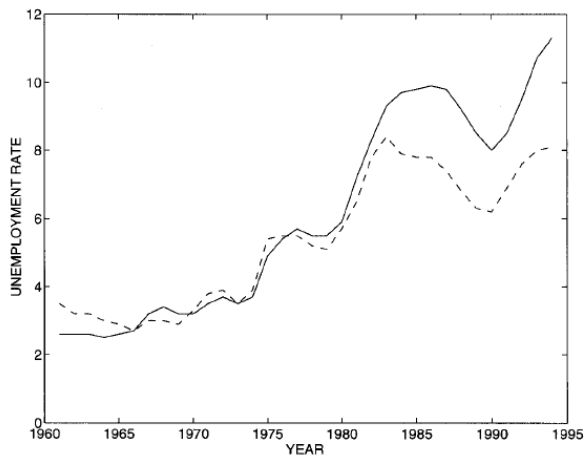
- Hypothesis: Everything that increases w_R (e.g. benefits b) will also increase the unemployment rate
- How does the unemployment rate relate to unemployment duration here?

Digression: micro-level evidence of model predictions I

- Following these insights, a large empirical literature has used micro data to investigate the effect of unemployment benefits on unemployment incidence, unemployment duration, and match quality
- Reservation wages and job-search behavior
 - ▶ Krueger-Mueller (AEJep 2016) use interview data to document self-perceived reservation wages
 - ▶ Marinescu-Skandalis (QJE 2020) use online application data to document how job-search behavior changes with benefit duration
- Unemployment duration and benefit generosity:
 - ▶ Older literature summarized in Krueger and Meyer (Handbook PE 2002)
 - ▶ Card-Lee-Pei-Weber (Econometrica, 2015) exploit kinks in Austrian UI
 - ▶ Kolsrud-Landais-Nilsson-Spinnewijn (AER 2017) exploit kinks in Swedish UI
 - ▶ Many more papers...
 - ▶ Raj Chetty's summary: 10 weeks of extra UI $\Rightarrow$ 1 week of extra unemployment (of course, the effect depends on all kinds of things)
 - ▶ Key input for evaluating optimal benefit provision: see, e.g., Chetty (JPubE, 2006), Chetty (JPE, 2008)

- Match quality and benefit generosity:
 - ▶ van Ours-Vodopivec (JLE 2006); Lalive (AER 2007); Card-Chetty-Weber (QJE 2007); Schmieder-von Wachter-Bender (AER 2016): zero effect
 - ▶ Nekoei-Weber (AER 2017) use Austrian micro data and an RD design: positive effect
- Lesson that appears in many of these papers: duration dependence matters
 - ▶ Reservation wage falls as unemployment benefit period is ending; human capital depreciates while unemployed
- Ljungqvist-Sargent (JPE 1998) argue that duration dependence is key for understanding differences in unemployment dynamics in Europe vs. the US

Digression: Ljungqvist-Sargent (JPE 1998)



Solid line: OECD Europe. Dashed line: OECD average.

Digression: Ljungqvist-Sargent (JPE 1998)

- Hypothesis: human capital depreciates over unemployment spell
 - ▶ Consistent with studies on mass layoffs: earnings losses after unemployment are very persistent (Lalonde-Jacobsen-Sullivan, AER 1993; Davis-von Wachter, BPEA 2012)
 - ▶ Evidence based on mass layoffs might confound partial- and general-equilibrium effects (Cederlöf 2019); see Lecture III on the DMP model
- In the 70s and 80s, European countries had generous replacement rates tied to previous wages, with close-to-unlimited duration
- Human capital depreciation + unlimited duration may result in $w_R > w_{offered} \Rightarrow$ long-term unemployment
- Makes little difference when macroeconomic volatility is low
- In the late 70s and 80s, macroeconomic volatility increased (oil price shocks) $\Rightarrow$ differences in UI institutions can explain why Europe, but not the US, got stuck in a high-unemployment equilibrium with substantial long-term unemployment

- Because of the randomness of wage offers, models of this sort generate **frictional wage dispersion**
- Hypothesis: some of the empirical wage dispersion that cannot be accounted for by observed heterogeneity is due to this kind of randomness
- Remember: Mincer regressions typically produce $R^2 < 0.3$ (Mortensen, Book 2003) and substantial residual dispersion: Acemoglu (JLE 2002) reports a 90/10 ratio ≈ 3.3 and a 50/10 ratio ≈ 1.8
- Can our model account for these numbers?

Frictional wage dispersion: how to test the model?

- To answer this question, the obvious approach seems to be
 - ① match the model parameters $\{r, \lambda_u, F(w), b\}$ to the data
 - ② solve for the reservation wage
 - ③ compute the wage distribution implied by the model $G(w) = F(w|w > w_R)$
 - ④ compare to the data
- Are there any problems with this approach?

Frictional wage dispersion: an alternative approach

- Hornstein-Krusell-Violante (AER 2011): use the information implicitly contained in households' search behavior
- Again, the reservation wage equation is:

$$w_R - b = \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - w_R) dF(w)$$

- The observed mean wage is $\bar{w} = E[w | w > w_R]$
- Rearrange

$$w_R - b = \frac{\lambda}{r + \sigma} \frac{\int_{w \geq w_R} (w - w_R) dF(w)}{1 - F(w_R)}$$

where $\lambda = \lambda_u(1 - F(w_R))$ is the observed job-finding rate

- What is the last ratio?

Frictional wage dispersion: a useful formula

- Step-by-step:

$$\begin{aligned}\frac{\int_{w \geq w_R} (w - w_R) dF(w)}{1 - F(w_R)} &= \frac{\int_{w \geq w_R} w dF(w)}{1 - F(w_R)} - w_R \frac{\int_{w \geq w_R} dF(w)}{1 - F(w_R)} \\ &= E[w | w > w_R] - w_R \frac{1 - F(w_R)}{1 - F(w_R)} \\ &= \bar{w} - w_R\end{aligned}$$

- and so the reservation equation can be written

$$w_R - b = \frac{\lambda}{r + \sigma} (\bar{w} - w_R)$$

- Write $b = \rho \bar{w}$, divide by w_R and rearrange

$$Mm = \frac{\bar{w}}{w_R} = \frac{\frac{\lambda}{r + \sigma} + 1}{\frac{\lambda}{r + \sigma} + \rho}$$

- Mm is a measure of wage dispersion and **the formula does not contain F !**
- Can the model match the data?
 - ▶ All parameters in our formula are observed in the data!
 - ▶ Plug in parameter values and compare the model-implied Mm ratio with the data

Frictional wage dispersion: testing the model

- US monthly data: $r = 0.0041$ (5% yearly interest rate), $\sigma = 0.03$, $\lambda = 0.43$, $\rho = 0.41$ (average replacement rate) $\Rightarrow$

$$Mm = \frac{\frac{\lambda}{r+\sigma} + 1}{\frac{\lambda}{r+\sigma} + \rho} = \frac{\frac{0.43}{0.0041+0.03} + 1}{\frac{0.43}{0.0041+0.03} + 0.41} = 1.046$$

- Reminder: 50/10-ratio in data ≈ 1.8
- The model comes nowhere near the data!**
- Intuition: high empirical λ puts tight constraint on the value of expected wage offer
 - ▶ High $\lambda \Rightarrow$ households must think it is not worth waiting long to get a better offer
 - ▶ $\Rightarrow$ expected wage from a new offer $\bar{w}$ cannot be much higher than outside option $\rho\bar{w}$
- The only ingredient is optimal household search behavior $\rightarrow$ a very robust result
- However, we shall see that one interesting avenue for thinking about frictional wage dispersion is models in which households also perform on-the-job search

- Key idea: unemployed households reject some wage offers because the option value of waiting for better offers dominates
- Generates predictions on unemployment rate, unemployment duration, match quality, wage dispersion
- Motivated by these predictions, a large empirical literature has emerged
- But the natural question is: where did the offer distribution F and contact rate λ_u come from?
 - ▶ Presumably vacancy/wage posting by firms?
- Also, we would presume that firm vacancy/wage posting behavior depends on worker search behavior, so can we really treat F and λ_u as exogenous?
 - ▶ We need a GE model to explore this!

General equilibrium? The Rothschild (JPE 1973) critique

- Suppose that the wage distribution F is generated by wage-posting firms with different productivities $p > b$
- Suppose that the firms know that all workers are identical and all workers follow the same reservation wage strategy (accept all $w > w_R$)
- What is the optimal wage posting rule of the firms?
- If posting $w > w_R$, you always get someone to work for you after a contact
- If posting $w < w_R$, you never get someone to work for you
- Optimal wage posting strategy is simply $w = w_R$, independent of p
- $\Rightarrow F$ is degenerate with unit mass point at $w = w_R$!

General equilibrium? The Diamond (JET 1971) paradox

- The problem goes deeper still. If there is no wage distribution, there is no point for an unemployed household to wait
- Our reservation wage equation:

$$\begin{aligned}w_R - b &= \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - w_R) dF(w) \\&= \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w_R - w_R) dF(w) \\&= 0\end{aligned}$$

- Ergo, optimal search behavior implies $w_R = b$ and all firms offer the monopoly contract $w = b$
- If workers have to pay a tiny fixed cost $\epsilon > 0$ for participating in search, nobody would search for jobs and there would be no production!
- Clearly, firm behavior and wage formation need to be addressed. Cliffhanger...

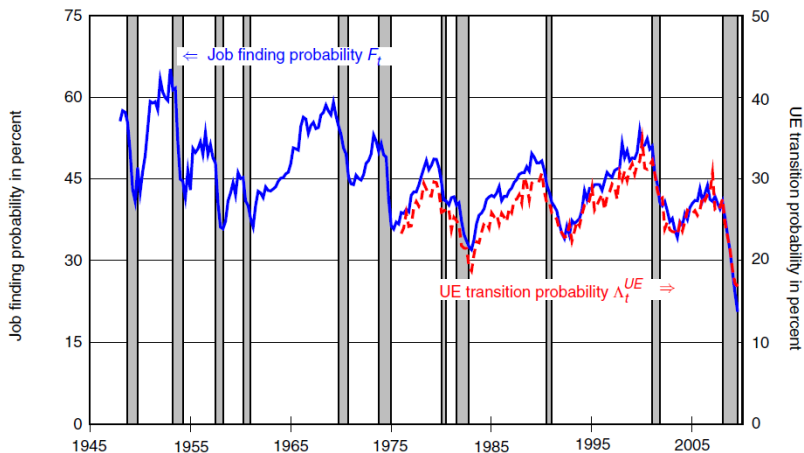
Bonus material: mapping data probabilities to model flow rates

- How to retrieve continuous-time transition rates from discrete-time transition probabilities (which are measurable in the data)?
- Our model says that in an interval $\delta t \rightarrow 0$
 - ① you can get at most one job
 - ② the probability of finding and accepting a job is $\lambda \delta t$
- Suppose we allowed workers to accept > 1 jobs within some empirically measurable time period Δt and that the arrival rate of job offers is independent of time
 - ▶ This does not change anything in our continuous-time model, as the probability of finding > 1 jobs in any given instant is negligible
- Then, the probability of finding k jobs in period Δt is $P(k|\Delta t)$, where P is given by the Poisson distribution with arrival rate λ
 - ▶ Hence, $P(k|\Delta t) = \frac{(\lambda \Delta t)^k}{k!} e^{-\lambda \Delta t}$
- What we can measure in the data is $F_t(\Delta t)$ — this is related to the continuous-time job-finding rate according to

$$F_t(\Delta t) = 1 - P(0|\Delta t) = 1 - e^{-\lambda \Delta t}$$

- If F_t is the monthly job-finding probability, then $\lambda = -\log(1 - F_t)$ is the monthly job-finding rate

Bonus material: mapping data probabilities to model flow rates



Blue solid line: inference from 2-state model; Red dashed line: inference from 3-state model. From Rogerson and Shimer (Handbook LE 2011).

- $F_t \approx 0.45 \Rightarrow \lambda_t \approx -\log(1 - 0.45) \approx 0.6$